EAMHP

Excel-Based Avionics Mechanical & Health Prediction

USER MANUAL

Complete Operating Guide — V1.1.0

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1. Overview

EAMHP (Excel-Based Avionics Maintenance & Health Prediction) is a Microsoft Excel-based engine health diagnostic tool designed for the Airbus A320 family. It enables Aviation Maintenance Engineers (AMEs), avionics trainees, and MRO personnel to perform structured engine fault detection by entering parameter readings and receiving instant, actionable maintenance guidance.

What This Manual Covers

This User Manual covers everything you need to operate EAMHP from first launch to fault output: system requirements, workbook orientation, step-by-step input instructions, output interpretation, a worked example, best usage tips, and key limitations. For logic internals, refer to EAMHP-LGM-001. For troubleshooting issues, refer to EAMHP-DBG-001.

EAMHP V1.1.0 covers:

- 680 engine fault combinations across 5 fault categories
- 2 engine variants: CFM56-5B and IAE V2500-A5 (Airbus A320 family)
- 16 engine performance parameters (N1, N2, EGT, Oil, Fuel Flow, Thrust, Starter)
- Corrective actions, preventive actions, and AMM/TSM procedures per fault
- Fully offline operation — no internet, no database, no external software required

2. System Requirements

EAMHP is a single-file Excel workbook. No installation is required. Ensure the following requirements are met before launching the tool.

Requirement	Minimum	Recommended	Notes
Microsoft Excel	Excel 2016 (v16.x)	Microsoft Excel 365	Must support VBA macros and XLOOKUP/VLOOKUP. Excel for Mac not guaranteed.
Operating System	Windows 10	Windows 10 / 11 (64-bit)	macOS compatibility not fully tested due to VBA differences.
File Format	.xlsm	.xlsm	Never save as .xlsx — this strips all VBA macros and disables the tool.
Macro Setting	Macros enabled	Enable all macros	Must be enabled in Excel Trust Center before use. See Section 3.
RAM	4 GB	8 GB or more	EAMHP V1.1.0 uses under 200 MB. More RAM

Requirement	Minimum	Recommended	Notes
			improves overall Excel responsiveness.
Screen Resolution	1280 × 720	1920 × 1080	Higher resolution ensures all sheet columns are visible without horizontal scrolling.

3. Getting Started

Follow these steps in order every time you launch EAMHP. Skipping Step 2 (macro activation) is the most common cause of the tool not functioning on first use.

STEP 1

Open the EAMHP Workbook

Locate the EAMHP workbook file on your computer and open it in Microsoft Excel.

- File name: EAMHP V1.1.0.xlsm
- Double-click the file to open it in Excel, or use File → Open within Excel
- Do NOT rename the file or change the .xlsm extension
- If opening from a USB drive or network folder, copy the file to your local hard drive (C:\) first for best performance

STEP 2

Enable Macros

EAMHP requires VBA macros to be active. Excel may block macros by default when opening a new file. You must enable them before the tool will work.

- If a yellow security bar appears at the top of the Excel window saying 'SECURITY WARNING: Macros have been disabled', click the 'Enable Content' button
- If no bar appears and macros are still blocked, go to: File → Options → Trust Center → Trust Center Settings → Macro Settings → select 'Disable all macros with notification' → click OK, then close and reopen the file
- To avoid this prompt every time, add the EAMHP folder to Excel Trusted Locations: Trust Center Settings → Trusted Locations → Add new location
- Verify macros are active: press Alt+F11 to open the VBA editor. If modules are visible with code inside, macros are enabled correctly

STEP 3

Navigate to the InputData Sheet

Once the workbook is open and macros are enabled, navigate to the primary working sheet for EAMHP.

- Click the 'InputData' tab at the bottom of the Excel window
- You will see column headers: TimeStamp, Engine Type, Parameter, Unit, Value, Upper Tolerance, Lower Tolerance, Fault Status
- The Upper Tolerance, Lower Tolerance, and Fault Status columns are auto-calculated — do not type into these columns

- If column widths appear too narrow, use Ctrl+A then double-click any column border to auto-fit all columns

STEP 4

Familiarise Yourself With the Workbook Sheets

EAMHP contains 6 sheets. Understanding each sheet's role will help you navigate the tool effectively.

- InputData — Your primary working sheet. Enter all parameter readings here.
- ReferenceData — Read-only tolerance reference. Do not edit this sheet.
- FaultMatrixEngine — Internal fault combination map. For reference only.
- DataValidation — Source of all dropdown lists. Do not edit this sheet.
- ProjectGuide — Built-in abbreviation glossary for AME terminology.
- Sheet2 — Reserved for future use and session archiving.

4. Workbook Sheet Guide

Each sheet in the EAMHP workbook has a defined role. This section describes what each sheet contains, what you can do with it, and what you must not modify.

📥 InputData — Primary Parameter Entry Sheet — USER EDITABLE

- Primary working area — enter all engine parameter readings here in separate rows
- Columns: TimeStamp (optional), Engine Type (dropdown), Parameter (dropdown), Unit (dropdown), Value (manual numeric entry)
- Fault Status, Upper Tolerance, and Lower Tolerance columns are formula-driven and auto-populate on value entry
- Add one row per parameter per maintenance session. Use a new session block for each new engine / maintenance event
- Safe to clear between sessions — archive completed sessions to Sheet2 before clearing

📚 ReferenceData — Tolerance Reference Database — READ-ONLY

- Contains certified tolerance records for all 16 Engine Performance Parameters (EPPs) for both CFM56-5B and IAE V2500-A5
- Columns: Parameter ID, Parameter Name, Engine Variant, Lower Tolerance, Upper Tolerance, Reference Value, Description, Unit
- Used automatically by VLOOKUP formulas in InputData to retrieve tolerance bounds
- DO NOT edit, sort, or delete any rows in this sheet — any modification will break tolerance lookups across InputData
- If a row appears incorrect, cross-verify against the current Airbus A320 AMM before considering any update

⚡ FaultMatrixEngine — Fault Combination Logic Reference — READ-ONLY

- Maps every 2-parameter and 3-parameter fault combination to a unique Fault_ID (F0001–F0680)
- Columns: Fault_ID, Parameters_Involved (comma-separated), State_Template (HIGH or LOW deviation pattern)
- Referenced by the Logic Engine to identify which Fault_ID corresponds to the current faulted parameter set
- DO NOT modify — any change to Fault_IDs or parameter strings will corrupt the fault identification pipeline
- For reference: open this sheet to manually look up whether a specific parameter combination exists in the fault matrix

📋 DataValidation — Dropdown Source Control — READ-ONLY

- Provides the source lists for all dropdown fields in InputData: Parameter Names (16 EPPs), Unit types (29 options), Engine Variants
- This sheet is the reason dropdown selections work correctly in InputData — it must not be modified
- If a parameter name or unit is missing from a dropdown, the issue originates here — restore from backup rather than manually adding entries
- Normally hidden from view during regular operation — navigate to it only for diagnostic purposes

📖 ProjectGuide — Built-In Glossary — REFERENCE

- Contains a numbered abbreviation glossary covering all aviation and EAMHP-specific terminology
- Includes entries for: EGT, N1, N2, ISA SL, TO/GA, ECAM, EFIS, GPS, EPP, FMS, and more
- Intended as a quick reference for AME trainees unfamiliar with specific aviation abbreviations
- Safe to read; no user data entry required on this sheet

5. Input Instructions

All engine parameter data is entered in the InputData sheet. Follow the field-by-field instructions below to ensure valid data entry. Incorrect input is the leading cause of wrong Fault Status results.

⚠️ Critical Input Rule

Always use the dropdown selectors for Engine Type, Parameter, and Unit fields. Do NOT free-type these values. Even a single extra space, capital letter difference, or abbreviation will cause a VLOOKUP failure and produce a #N/A Fault Status instead of a valid evaluation.

Field Name	Input Type	Required	Valid Values / Source	Important Notes
TimeStamp	Free entry (Date/Time)	**OPTIONAL**	Any date-time format	Used for session audit trail. Leave blank if not needed.
Engine Type	Dropdown	**YES**	CFM56-5B | IAE V2500-A5	Must match actual engine on aircraft. Determines which tolerance band is used.
Parameter	Dropdown	**YES**	16 EPPs (see Section 5.1)	Select exact parameter name. Partial matches cause #N/A.
Unit	Dropdown	**YES**	29 unit types (see Sec 5.2)	Select unit matching your measurement instrument. Unit is recorded but does not affect tolerance math.
Value	Numeric (manual)	**YES**	Any numeric value	Enter the raw numeric reading. Do NOT include units, text, or symbols in this field.
Upper Tolerance	AUTO (formula)	**AUTO**	Retrieved from ReferenceData	Do not type in this field. Auto-populates on Engine Type + Parameter entry.
Lower Tolerance	AUTO (formula)	**AUTO**	Retrieved from ReferenceData	Do not type in this field. Auto-populates on Engine Type + Parameter entry.
Fault Status	AUTO (formula)	**AUTO**	FAULT / NORMAL / #N/A	Result of Value vs. tolerance evaluation. FAULT = deviation detected. #N/A = invalid input.

5.1 Supported Engine Performance Parameters (EPPs)

The following 16 parameters are available in the Parameter dropdown. Enter readings only for parameters you have measured. Do not enter estimated or assumed values.

#	Parameter Name	Engine Variant	Unit	Typical Range
1	N1 Idle	CFM56-5B	%	18.1 – 24.2% (ref: ~19.5–23%)
2	N1 Idle	IAE V2500-A5	%	19.0 – 21.0% (ref: ~20%)
3	N1 Max	Both variants	%	98.8 – 109.2% (TO/GA ref: 104%)
4	N2 Idle	CFM56-5B	%	55.1 – 65.1% (ref: 58–62%)
5	N2 Idle	IAE V2500-A5	%	57.0 – 63.0% (ref: ~60%)
6	N2 Max	Both variants	%	99.8 – 110.3% (ref: ~105%)
7	EGT Idle	CFM56-5B	°C	370.5 – 456.8°C (ref: 390–435°C)
8	EGT Idle	IAE V2500-A5	°C	380.0 – 472.5°C (ref: 400–450°C)

#	Parameter Name	Engine Variant	Unit	Typical Range
9	EGT Max (TO/GA)	CFM56-5B	°C	603.3 – 666.8°C (ref: 635°C / 5 min)
10	EGT Max (TO/GA)	IAE V2500-A5	°C	617.5 – 682.5°C (ref: 650°C / 5 min)
11	EGT Max Continuous	Both variants	°C	579.5 – 640.5°C (ref: 610°C)
12	Oil Pressure Min	Both variants	psi	13.0 – 75.0 psi
13	Oil Temp Max Continuous	Both variants	°C	Per engine AMM limit
14	Oil Quantity Min	Both variants	qt/L	6.5 – 22.0 qt
15	Fuel Flow Cruise	Both variants	kg/h	Per cruise flight level and thrust setting
16	Starter Limit	Both variants	sec	Per AMM start duty cycle limits

6. Running the System

EAMHP is primarily formula-driven, meaning fault evaluation begins automatically as soon as you enter a Value in the InputData sheet. No separate 'Run' button is required for basic fault status evaluation.

> ⚡ **Automatic Evaluation**
>
> As soon as you enter a numeric Value in the InputData Value column, Excel automatically retrieves the tolerance bounds and evaluates the Fault Status. There is no need to press Enter or click a button — the result appears immediately in the Fault Status column.

6.1 Basic Operation Workflow

Step	Action	Expected Result
1	Click on the Engine Type cell in the InputData row and select engine variant from dropdown	Engine Type populated. Upper/Lower Tolerance columns may show #N/A until Parameter is also selected.
2	Click on the Parameter cell and select the parameter name from the dropdown	Parameter populated. If Engine Type is already selected, Upper and Lower Tolerance values auto-populate from ReferenceData.
3	Click on the Unit cell and select the appropriate engineering unit from the dropdown	Unit recorded. Does not affect tolerance evaluation but is important for documentation accuracy.
4	Click on the Value cell and type the measured numeric reading, then press Tab or Enter	Fault Status column immediately shows FAULT or NORMAL based on the value vs. tolerance comparison.

Step	Action	Expected Result
5	Repeat rows 1–4 for each additional parameter in the current maintenance session	Multiple Fault Status results build up, enabling multi-parameter fault correlation in the Logic Engine.
6	Navigate to the relevant category sheet (e.g., Oil_System_Malfunction) or Mixed Report sheet to view the full maintenance output for identified Fault_IDs	Complete fault record displayed: Fault_ID, Category, Corrective Action, Preventive Action, CFM56-5B and V2500 procedures.

7. Understanding the Output

Once the Logic Engine identifies a Fault_ID from the faulted parameter combination, the full maintenance output is available in the fault matrix sheets. Each output record contains six fields covering the complete maintenance decision.

	Output Field	Description	Example Value
	Fault_ID	Unique identifier for the matched fault combination	*F0007*
	Parameters_Involved	Comma-separated list of all parameters that triggered this fault	*N1 Idle, Oil Pressure Min*
	Fault_Category	Classified fault type driving the maintenance decision pathway	*Oil System Malfunction*
	Corrective_Action	Immediate maintenance action to be performed by the AME	*Inspect oil pump, check filter and cooler, verify oil qty and pressure*
	Preventive_Action	Scheduled maintenance recommendation to prevent fault recurrence	*Regular oil analysis, scheduled filter replacement, periodic oil system inspection*
	CFM56-5B Procedure	Engine-specific AMM/TSM procedure for CFM56-5B variant	*Inspect oil pump, filter, and cooler; check scavenge system and leaks*
	V2500 Procedure	Engine-specific AMM/TSM procedure for IAE V2500-A5 variant	*Inspect lubrication pump, filter bypass valve, verify oil cooler and check for leaks*

7.1 Fault Status Indicators

Fault Status	Meaning	Action Required
NORMAL	Parameter value is within the certified tolerance band. No fault detected for this parameter.	No immediate action. Continue monitoring.

Fault Status	Meaning	Action Required
FAULT	Parameter value is outside tolerance (HIGH or LOW). Fault condition detected.	Note the parameter and enter all related parameters to build the multi-parameter fault signature for correlation.
#N/A	Lookup failed. Parameter Name or Engine Type does not match any ReferenceData record.	Reselect Parameter and Engine Type from dropdowns. Do not free-type these fields.
[Blank]	Value field has not been entered yet. Evaluation has not run.	Enter a numeric Value to trigger evaluation.

8. Worked Example — End-to-End Fault Identification

This section walks through a complete EAMHP session from data entry to maintenance output using a realistic A320 engine fault scenario.

📋 **Scenario**

During a line maintenance check on an Airbus A320 with a CFM56-5B engine, the AME records the following readings: N1 Idle = 17.0% (below normal range) and Oil Pressure Min = 11.0 psi (below minimum). Both parameters are outside their certified limits. Enter these into EAMHP and identify the fault.

8.1 Data Entry

Row	Engine Type	Parameter	Unit	Value Entered	Lower Tol	Upper Tol	Fault Status
1	CFM56-5B	N1 Idle	%	17.0	18.1	24.2	FAULT (LOW)
2	CFM56-5B	Oil Pressure Min	psi	11.0	13.0	75.0	FAULT (LOW)

8.2 Logic Engine Processing

- Both parameters show FAULT status → the Logic Engine detects a 2-parameter fault condition
- Parameter combination: [N1 Idle + Oil Pressure Min] is cross-referenced against FaultMatrixEngine
- Match found: Fault_ID = F0007
- Fault_Category = Oil System Malfunction

8.3 Output Retrieved

Output Field	Value for F0007
Fault_ID	F0007

Output Field	Value for F0007
Parameters_Involved	N1 Idle, Oil Pressure Min
Fault_Category	Oil System Malfunction
Corrective_Action	Inspect oil pump, check oil filter and cooler, inspect for leaks, verify oil quantity and pressure.
Preventive_Action	Regular oil analysis, scheduled oil filter replacement, periodic oil system inspection.
CFM56_5B_Procedure	CFM56-5B: Inspect oil pump, oil filter, and oil cooler; check scavenge system and leaks.
V2500_Procedure	V2500: Inspect lubrication pump, oil filter bypass valve, verify oil cooler performance and check for leaks.

✅ Outcome

EAMHP correctly identified an Oil System Malfunction from the combination of low N1 speed and low oil pressure, consistent with a failing oil pump scenario. The AME now has a specific corrective action and the CFM56-5B AMM/TSM procedure reference to proceed with inspection. Cross-verify against the current AMM before performing any maintenance action.

9. Best Usage Tips

These tips will help you get the most accurate, reliable results from EAMHP and develop strong diagnostic habits aligned with real-world AME practice.

	Tip	Details
📏	**Use Realistic Values**	Always enter readings taken from actual aircraft instruments, maintenance records, or certified training data. Entering guessed or random values will produce meaningless outputs and poor learning outcomes.
📊	**Enter All Faulted Parameters**	For the most accurate fault identification, enter ALL parameters you have measured in a single session — not just the obvious fault. EAMHP's logic is designed to identify compound 3-parameter fault signatures that single-parameter evaluation would miss.
✈️	**Match Engine Type Exactly**	Confirm the engine variant on the specific aircraft before selecting Engine Type. CFM56-5B and V2500-A5 have different tolerance bands — selecting the wrong engine produces incorrect Fault Status results and wrong AMM procedures.
📚	**Cross-Check AMM**	Always verify EAMHP outputs against the current revision of the Airbus A320 AMM before taking any maintenance action. AMM revisions may update tolerance values that are not yet reflected in EAMHP V1.1.0.
📝	**Archive Sessions**	Before clearing InputData for a new session, copy the data rows to Sheet2 with a date/session label for your records. This helps you track diagnostic patterns and supports training documentation.

	Tip	Details
🧪	**Use for Training Scenarios**	EAMHP is excellent for scenario-based learning. Try entering different parameter combinations to explore fault categories, understand which parameter deviations lead to which maintenance actions, and build your diagnostic pattern recognition.
🔄	**Repeat for 3-Parameter Faults**	If you have three simultaneously faulted parameters, enter all three in the same InputData session. The Logic Engine can then identify 3-parameter Fault_IDs (F0121–F0680) which represent more complex and clinically important fault signatures.
⚠️	**Training Tool — Not Airworthy**	EAMHP is a decision-support and training tool. It is not an approved airworthiness system. It must not be used as the sole basis for an airworthiness release or maintenance sign-off on an actual aircraft.

10. Limitations

Understanding what EAMHP cannot do is as important as knowing what it can do. The following limitations apply to V1.1.0.

Limitation	Description	Workaround / Future Plan
Training & support use only	EAMHP is not an approved airworthiness tool and cannot be used for regulatory maintenance sign-off	Always cross-reference with official AMM. For certification: consult EASA Part-66 / DGCA resources.
Manual data entry	All parameter readings must be entered manually by the user. No ACARS, QAR, or sensor data feed exists.	V1.2.0 roadmap: ACARS data import module planned.
16 parameters in V1.1.0	Vibration, hydraulic, fuel flow idle, and avionics LRU parameters are not yet included.	V1.0.1: Expanded parameter set planned.
Max 3 simultaneous faults	EAMHP handles up to 3 concurrent faulted parameters. 4+ concurrent deviations are not covered.	V1.0.1: 4-parameter combination expansion planned.
CFM56-5B & V2500 only	Only the A320ceo engine variants are covered. A320neo (LEAP-1A, PW1100G) are not yet supported.	V1.1.0: A320neo engine variant expansion planned.
ISA SL conditions assumed	Reference tolerances are based on ISA Standard Atmosphere / Sea Level. Non-standard conditions may affect fault accuracy.	Apply ISA deviation corrections before entering values where required.
Windows Excel only	macOS Excel and Google Sheets are not supported due to VBA incompatibilities.	Long-term: Web application version under evaluation.